

Trainer Debrief — Lab 20

Multi-Vari Studies and Chi-Square Analysis (Black Belt Only)

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Decompose variation by family using a multi-vari study and test categorical association (A3, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-20-multivari-chisquare.xlsx

MultiVari: the LINE-TO-LINE (cyclical) family dominates — L3 sits ~9 kPa below L1 and L2, while cavity-to-cavity (positional) spans only ~3 kPa and the time-to-time (temporal) drift is ~1 kPa across the shift. Focus the investigation on what is different about Line 3; stop sampling the other two families. ChiSquare: chi-sq ~ 21.3, df = 6, $p < 0.01$ — defect MODE is not independent of plant. Suzhou's excess is concentrated in seal weld leaks specifically, which points back at the fixture, not at a general quality problem.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Your multi-vari chart captures all three families with quantified variance contributions, your cause list is re-prioritised against the dominant family, and your chi-square satisfies the expected-frequency condition.

Where this sits in the capstone

BLACK BELT ONLY. Before spending money hunting a root cause, find out WHERE the variation actually lives. A multi-vari study separates positional, cyclical and temporal variation — and frequently reveals that the team has been investigating entirely the wrong family. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd · TRAINER ONLY